One-Sided, Infinite, and End-Behavior Limits

One-sided limits, infinite limits at a vertical asymptote, end behavior at infinity, and continuity of a piecewise function. Decide which of the three situations you are in before computing:

- **One-sided.** The superscript $+$ or $-$ tells you which branch of the function to use and which sign a small factor carries.
- **Infinite.** A nonzero numerator over a denominator going to zero is $\pm\infty$; get the sign from the sign of the denominator on that side. Write ∞ or $-\infty$, or "does not exist" with a reason — never a number.
- **End behavior.** As $x \to \pm\infty$, compare degrees, or divide numerator and denominator by the highest power of x in the denominator.

Show the reasoning line that decides the answer, not just the answer.

1. Evaluate: $\displaystyle \lim_{x\to 2^+} \frac{|x-2|}{x-2}$

Answer: ____________________

2. Evaluate: $\displaystyle \lim_{x\to \infty} \frac{3x^2+5x-1}{x^2-4}$

Answer: ____________________

3. Evaluate: $\displaystyle\lim_{x \to 5^-} \frac{1}{x-5}$

State the sign of $x - 5$ for values just below 5 as part of your reasoning.

Answer: ____________________

4. Evaluate: $\displaystyle\lim_{x \to -\infty} \frac{4x^3 - x}{2x^3 + 7}$

Answer: ____________________

5. Let

$$f(x) = \begin{cases} x^2 + 1, & x < 2 \\ 5x - 3, & x \geq 2 \end{cases}$$

(a) Find $\displaystyle\lim_{x \to 2^-} f(x)$. (b) Find $\displaystyle\lim_{x \to 2^+} f(x)$. (c) Does $\displaystyle\lim_{x \to 2} f(x)$ exist? Explain in one sentence.

(a) ____________ (b) ____________

6. Evaluate: $\displaystyle\lim_{x\to -4^+} \frac{2x}{x+4}$

Answer: ________________

7. Evaluate: $\displaystyle\lim_{x\to\infty} \frac{\sqrt{9x^2+4}}{2x+1}$

Divide by the highest power of x in the denominator, and be explicit about what $\sqrt{x^2}$ equals when x is large and positive.

Answer: ________________

8. Evaluate: $\displaystyle\lim_{x\to 0^+} \left(\frac{1}{x} - \frac{1}{x^2}\right)$

Combine over a common denominator first — the two pieces separately are both infinite, and their difference is not settled until they are combined.

Answer: ________________

9. Let

$$g(x) = \begin{cases} x^2 + k, & x < 3 \\ 4x - 1, & x \geq 3 \end{cases}$$

Find the value of the constant k that makes g continuous at $x = 3$. Write the one-sided limits first, set them equal, and solve.

Answer: ________________

10. **Synthesis challenge.** Let $h(x) = \dfrac{2x^2 - 3x}{x^2 - x - 6}$.

(a) Find $\lim_{x\to\infty} h(x)$. (b) Find $\lim_{x\to 3^-} h(x)$. (c) Find $\lim_{x\to 3^+} h(x)$. (d) Using your answers, describe every vertical and horizontal asymptote of the graph of h in words.

(a) __________ (b) __________ (c) __________